Gene Pulser® Electroprotocols

*** We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.**

Cell Type Mammalian, adherent

Species Used Monkey, COS-7, kidney, SV-40 transformed

Molecules Electroporated DNA: CMV b-gal , 6 kB, supercoiled

Before the Pulse

Cell Growth Medium DMEM + 10% Fetal Bovine Serum (GIBCO/BRL, Sigma)

Growth Phase at Harvest Log phase, 70 to 80% confluent

Pre-pulse Incubation Room temperature

Wash Solution Trypsinize

The Pulse

Instruments Used Gene Pulser® apparatus & Capacitance

Electroporation Temperature Room temperature

Electroporation Medium* DMEM + 10% Fetal Bovine Serum + 5mM BES, pH 7.2

Cell Density 2 x 10 (7) cells / ml

Volume of Cells 10 µg / 250 µl DNA

DNA Concentration 10 mM Tris, 1 mM EDTA, pH 8.0)

DNA Resuspension Buffer Not given

Volume of DNA Not given

Cuvette Gap 0.4 cm

Voltage 0.170 kV

Field Strength 0.425 kV/cm

Capacitor 960 µF

Resistor (Pulse Controller) Ω none

Time Constant 45 msec

After the Pulse

Outgrowth Medium Not given

Outgrowth Temperature Room temperature

Length of Incubation 10 min., then pellet& resuspend in DMEM

Selection Method or Assay Used Not given

Electroporation Efficiency 25%

Per Cent Survival about 100%

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

See reference: Ustav, M., and Stenlund, A. 1991. *EMBO J*. **10**(2):449-457.

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